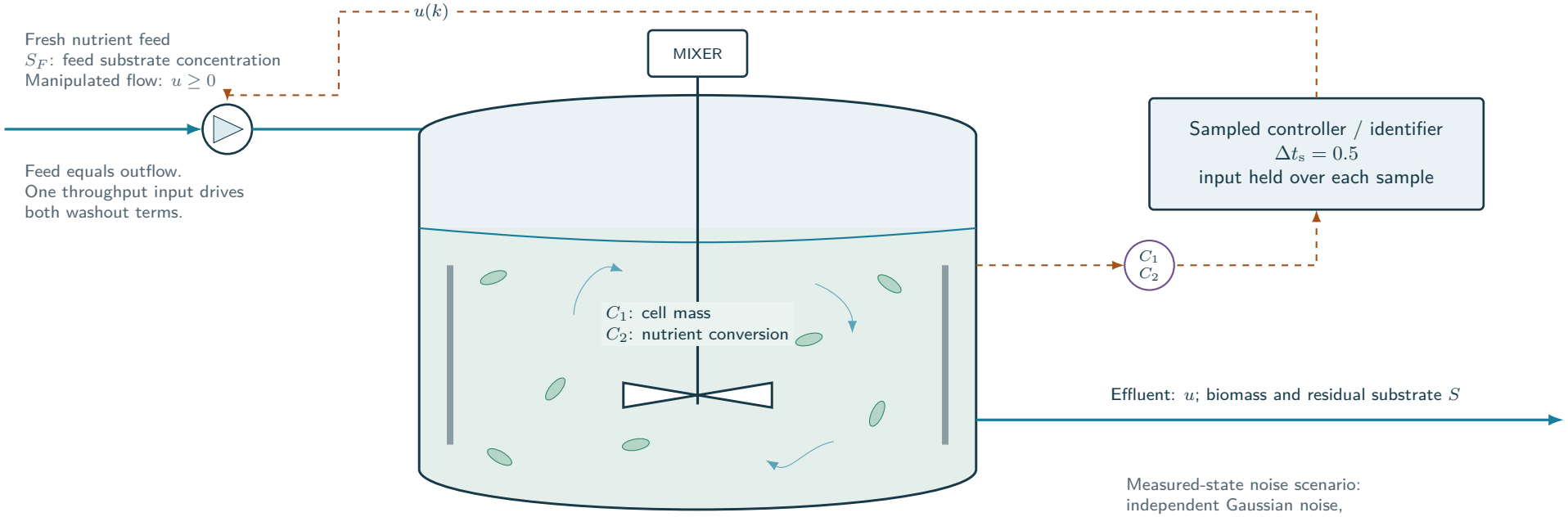
Continuous biochemical reactor

Paper section 3.8: *Continuously operated bio-chemical reactor*

Two nonlinear states | Constant-volume, well-mixed process | Dimensionless model and time



Normalized state definitions

$C_2 = (S_F - S)/S_F, \quad 0 \leq C_1, C_2 \leq 1.$

$r = C_1(1 - C_2)\exp(C_2/\Gamma).$

Nominal $\Gamma = 0.4$, $\beta = 0.02$.

Growth and washout balances

$$\dot{C}_1 = -uC_1 + r$$

$$\dot{C}_2 = -uC_2 + r\frac{1+\beta}{1+\beta-C_2}.$$

No dimensional flow or time scale is supplied.

What makes this a benchmark

Stable and unstable operating regions; limit cycles and a non-unique input–yield relation.

One sampled transition uses **50 Euler substeps**, each of length 0.01.

Scope: Eqs. (37)–(38), with dimensionless throughput u. The mixing hardware is explanatory. Equilibria must be recomputed from the equations; the printed biomass at $u = 0.769$ is not consistent with Eq. (37).

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.